The Quantum Structure Sub-Programme

Presentation Note 7

Jérôme Beau

Independent Researcher

jerome.beau@cosmochrony.org

2026

Abstract

The quantum structure sub-programme of the Cosmochrony corpus addresses a single central question: how does the admissible SU(2) sector force the structures of quantum mechanics, without importing quantum postulates? Starting from the admissibility constraints of A1–A4 on the binary icosahedral group $2I$ and the Weil representation V_ρ, the sub-programme derives: phase coherence of the admissible fibre (Q1); the singlet correlator $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$ (Q1, Q3); the Tsirelson bound $|S_{\mathrm{CHSH}}| \leq 2\sqrt{2}$ (Q1); the Born rule in the SU(2) sector (Q1, Q3); the identification of SU(2) as the unique stable fixed point of the admissibility flow (Q2); the universal spin-j generalisation of all the above for all five admissible sectors $j \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}$ of $2I$ (Q3); and the structural non-applicability of Bell factorizability in non-injective frameworks (Bell paper). All these results are proved within the SU(2) sector. Extension to arbitrary observables, multipartite systems, and continuous spectra remains the primary open deliverable.

Keywords. quantum structure; admissibility; phase coherence; Born rule; singlet correlator; Tsirelson bound; Bell non-applicability; SU(2); binary icosahedral group; non-injective projection; presentation note.

Contents

1 Motivation and Central Question **3**

2 Position in the Cosmochrony Programme **3**

3 Logical Chain of the Sub-Programme **4**

4 Constituent Papers **4**
4.1 Phase coherence and the spin-1/2 quantum sector: Q1 4
4.2 Co-admissible sectors and the SU(2) fixed point: Q2 5
4.3 Universal spin-j quantum sector: Q3 5
4.4 Bell non-applicability: Bell paper 5

5 Inputs from Upstream Results **6**

6 Outputs to Downstream Branches **6**

7 Status of Results **6**
7.1 Proved results 6
7.2 Numerical evidence 7
7.3 Open hypotheses 8

8 Open Deliverables **8**

9 Conclusion **8**

1 Motivation and Central Question

The quantum structure sub-programme answers the following question.

Central question. The admissibility axioms A1–A4 force the fibre $F_n \simeq V_\rho$ with Heisenberg structure (Note 5). They also force the admissibility thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$, with the spin-$\frac{1}{2}$ sector identified as the physically admissible sector (Note 1). *Do the structures of quantum mechanics — complex amplitudes, Born rule, singlet correlator, Bell-type correlations — follow from admissibility alone, or must they be separately postulated?*

The sub-programme answers: they follow. Phase coherence is forced by the BI indiscernibility of conjugate Weil blocks, which is itself a consequence of the admissibility constraint (Q1). The singlet correlator $E = -\hat{a} \cdot \hat{b}$ follows from four independently established inputs: linearity of $\mathfrak{su}(2)$ observables (O23), bilinearity from coherent amplitude structure (Q1), rotational invariance from isotropic saturation (O23), and normalisation from the parity involution (O18). No quantum postulate is invoked at any step.

This sub-programme is the "first physics" of the corpus: before spacetime geometry, before gauge structure, before gravity, the admissibility constraints already force quantum correlations.

Remark 1 (Vocabulary note)**.** Throughout this note, admissibility replaces the earlier relaxation vocabulary. The substrate χ is static. Quantum structure is not the dynamics of χ but the algebraic consequence of non-injective admissible transitions.

Remark 2 (Scope: SU(2) sector only)**.** All results in this sub-programme are proved within the SU(2) admissible sector on the binary icosahedral group $2I$. Extension to arbitrary observables and multipartite systems requires extending the fibre structure beyond the two-element parity sector of O18 and remains open.

2 Position in the Cosmochrony Programme

The quantum structure sub-programme occupies an early and independent position in Branch III. It depends on Branch I (axioms, $F_n \simeq V_\rho$, BI parity) and on selected O-series results (O18, O22, O23), but not on the geometry, gauge, or gravity sub-programmes. This logical independence is important: quantum structure is not derived from spacetime; it is prior to it.

Inputs:

- Admissible fibre $F_n \simeq V_\rho$ and Heisenberg structure (Note 5, Foundation [1], HeisenbergStructure [6]).
- Admissibility thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$; spin-$\frac{1}{2}$ as minimal admissible sector (Note 1, SpectralAdmissibility [10]).
- $\mathrm{Im}\,\mathbb{H} \cong \mathfrak{su}(2)$; three isotropic directions; isotropy condition $M \propto I$ (O23 [11]).
- Parity involution $c \leftrightarrow q - c$; unit-norm amplitudes at saturation (O18 [5]; O22 [9]).
- BI indiscernibility of conjugate Weil blocks (BornInfeld [3]).

Outputs feed into:

- Note 2 (Emergent Geometry): SU(2) as the admissible sector motivates the identification $H_{\text{eff}} = \mathrm{Sym}^2(V_\rho)$.
- Note 3 (Gauge Structure): the phase coherence and SU(2) fixed-point identification underpin the gauge-group derivation.
- Note 6 (Fermionic Matter): the Born rule and coherent amplitude structure are assumed in the spinorial sector construction.

3 Logical Chain of the Sub-Programme

$$\underbrace{F_n \simeq V_\rho,\ \text{BI indisc.}}_{\text{A1–A4, O18}} \implies \underbrace{\text{phase coherence}}_{\text{Q1 Thm 2.7}} \implies \underbrace{E(\hat{a},\hat{b}) = -\hat{a}\cdot\hat{b}}_{\text{Q1 Thm 2.14}} \implies \underbrace{|S_{\text{CHSH}}| \le 2\sqrt{2}}_{\text{Q1 Cor 2.16}} \implies \underbrace{\text{Born rule}}_{\text{Q1 Thm 2.18}}$$

$$\implies \underbrace{\text{SU(2) fixed point}}_{\text{Q2 Thm 8.6}} \implies \underbrace{E = -\tfrac{j(j+1)}{3}\hat{a}\cdot\hat{b},\ \text{all } j}_{\text{Q3 Thm 5.2}} \tag{1}$$

Four structural steps.

1. *Phase coherence from BI indiscernibility (Q1).* Any admissible transition must preserve the BI indiscernibility of conjugate Weil blocks ρ_c and ρ_{q-c}. Destroying this indiscernibility would inflate the rank of the combined Gram–Schmidt span, violating the admissible rank growth of O22. Phase coherence is therefore a structural consequence of admissibility, not a postulate.

2. *Singlet correlator from four structural inputs (Q1).* Bilinearity (from coherent amplitudes, step 1), linearity of $\mathfrak{su}(2)$ observables (O23), rotation invariance (isotropic saturation, O23), and unit normalisation (O18 parity) together force $E(\hat{a},\hat{b}) = -\hat{a}\cdot\hat{b}$. No quantum postulate is invoked.

3. *Tsirelson bound as corollary (Q1).* Since the correlator is derived rather than assumed, the Cauchy–Schwarz bound on the CHSH combination gives $|S_{\text{CHSH}}| \le 2\sqrt{2}$ unconditionally.

4. *Born rule from structural uniqueness (Q1).* The unique probability assignment compatible with positivity and normalisation, linearity of mean values (from coherent amplitudes), and the derived correlator is $P(a = +1|\hat{a}) = |\langle +\hat{a}|\psi\rangle|^2$. No Gleason theorem is needed.

5. SU(2) *as stable fixed point (Q2).* The spin-$\frac{1}{2}$ and spin-$\frac{3}{2}$ sectors of $2I$ share the Laplacian eigenvalue $\lambda_{1/2} = \lambda_{3/2} = 18$ (co-admissibility). The admissibility flow — the sequence of spectral refinements as $p \to \infty$ in the LPS approximation — selects $j = \frac{1}{2}$ as the unique stable fixed point. All larger-j sectors are unstable under the flow.

6. *Universal spin-j generalisation (Q3).* For all five admissible sectors $j \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}$ of $2I$: the proto-state is the singlet $|\Omega_j\rangle$ (forced by Schur's lemma on the Clebsch–Gordan decomposition); the universal correlator is $E(\hat{a},\hat{b}) = -\frac{j(j+1)}{3}(\hat{a}\cdot\hat{b})$; and the Born rule holds in each sector.

4 Constituent Papers

4.1 Phase coherence and the spin-1/2 quantum sector: Q1

Q1 [4] is the foundational paper of the sub-programme. It establishes the structural derivation of quantum mechanics within the spin-$\frac{1}{2}$ sector.

Phase coherence (Theorem 2.7; proved). Any admissible transition preserves BI indiscernibility of conjugate Weil blocks ρ_c and ρ_{q-c}, thereby maintaining the metaplectic phase coherence of the fibre. Phase coherence is not postulated; it is the unique behaviour compatible with the admissible rank growth of O22. *Observable signature*: any coherence-destroying transition inflates the rank of the Gram–Schmidt span of $\{\rho_c, \rho_{q-c}\}$ beyond the O22-admissible bound (Corollary 2.9; falsifiable). Numerically confirmed for $q = 29$: rank $W_n^{(c)} = 0$ exactly throughout the admissible regime across all four conjugate pairs.

Singlet correlator (Theorem 2.14; proved). $E(\hat{a},\hat{b}) = -\hat{a}\cdot\hat{b}$ from the four inputs: O23 ($\operatorname{Im}\mathbb{H} \cong \mathfrak{su}(2)$ linear observables), Theorem 2.7 (bilinear evaluation from coherent amplitudes), O23 (isotropic saturation $M \propto I$), O18 (normalisation from parity involution).

Tsirelson bound (Corollary 2.16; proved unconditionally). $|S_{\text{CHSH}}| \le 2\sqrt{2}$; this is now a structural consequence of the geometry of $\operatorname{Im}\mathbb{H}$ under admissible projection, not an import from quantum mechanics.

Born rule (Theorem 2.18; proved). $P(a = +1|\hat{a}) = |\langle +\hat{a}|\psi\rangle|^2$ is the unique probability assignment compatible with the coherent amplitude structure and the derived correlator. Uniqueness follows from the structural constraints; no Gleason theorem is needed.

4.2 Co-admissible sectors and the SU(2) fixed point: Q2

Q2 [8] extends the programme to the spin-$\frac{3}{2}$ sector of $2I$ and identifies SU(2) as the unique stable fixed point.

Co-admissibility of spin-$\frac{1}{2}$ and spin-$\frac{3}{2}$ (proved). The two sectors share the Laplacian eigenvalue $\lambda_{1/2} = \lambda_{3/2} = 18$ on $2I$, making them simultaneously compatible with the admissibility envelope.

Quantum structure for spin-$\frac{3}{2}$ (proved). Phase coherence, the singlet correlator $E = -\frac{5}{4}\hat{a}\cdot\hat{b}$, the Tsirelson-type bound, and the Born rule are derived for the χ_4 sector by the same structural arguments as Q1.

SU(2) *as stable fixed point* (Theorem 8.6; proved). The admissibility flow selects $j = \frac{1}{2}$ as the unique stable sector in the large-p LPS limit: all co-admissible $j > \frac{1}{2}$ sectors are unstable under spectral refinement. This recovers the standard correlator $E = -\hat{a}\cdot\hat{b}$ and the standard Tsirelson bound $2\sqrt{2}$ as the physically realised values.

4.3 Universal spin-j quantum sector: Q3

Q3 [12] completes the SU(2) quantum sector for all five admissible sectors of $2I$.

Proto-state is the singlet, all j (Theorem 4.1; proved). BI indiscernibility forces the proto-state to be invariant under the diagonal $2I$-action on $V_j \otimes V_j$. By Schur's lemma on the Clebsch–Gordan decomposition $\chi_{2j+1}\otimes\chi_{2j+1} = \bigoplus_{J=0}^{2j}\chi_{2J+1}$, the unique normalised invariant vector is the singlet $|\Omega_j\rangle$. This closes the open identification step of Q2 for spin-$\frac{3}{2}$ and establishes the result for all j.

Universal correlator (Theorem 5.2; proved). For any admissible sector χ_{2j+1} of $2I$: $E(\hat{a},\hat{b}) = -\frac{j(j+1)}{3}(\hat{a}\cdot\hat{b})$. The standard correlator is recovered at $j = \frac{1}{2}$ where the normalised prefactor $4\cdot j(j+1)/3 = 1$ (Table 2 of Q3).

Born rule, all j (Corollary 6.1; proved). The unique probability assignment for any admissible sector χ_{2j+1} is $P(m|\hat{a}) = |\langle m|\psi\rangle|^2$, following from the same structural argument as Q1 Theorem 2.18 applied sectorwise.

Closure of the SU(2) *quantum sector* (Remark 7.1). Combining Q1, Q2, and Q3: (i) phase coherence derived; (ii) proto-state is the singlet; (iii) universal correlator derived for all j; (iv) Born rule holds in each sector. The admissibility flow of Q2 (Theorem 8.6) selects $j = \frac{1}{2}$ as the physical sector in the continuum limit.

4.4 Bell non-applicability: Bell paper

Bell paper [2] is published in *Quantum Reports* (MDPI, 2026) and is the only published result of the sub-programme. It establishes that Bell-type factorizability does not apply in non-injective effective descriptions.

Non-applicability of Bell factorizability (proved). In any framework where observable outcomes arise as equivalence classes of underlying configurations under a non-injective map, the standard probabilistic factorization assumption implicit in Bell's theorem does not hold. The proof is structural: it requires only the non-injectivity of Π and the informational completeness condition (E2 of ENI [7]). No hidden variable, no non-local dynamics, and no modification of quantum mechanics is introduced.

Mechanism: observable outcomes correspond to equivalence classes $\Pi^{-1}(O_n)$; distinct configurations within the same class cannot be assigned independent local variables, preventing the joint factorization $p(a,b|\hat{a},\hat{b}) = p(a|\hat{a},\lambda)\cdot p(b|\hat{b},\lambda)$ for any common λ.

Toy model: an explicit model produces Bell–CHSH violations while preserving operational no-signalling and statistical independence of measurement settings. The model may exceed the Tsirelson bound (its role is to isolate the structural mechanism, not to reproduce quantum correlations quantitatively).

Operator reformulation: in the operator language, the same mechanism reformulates as a reduction of the observable algebra to a non-commutative conditional expectation.

Table 1: Summary of results by paper. Status: P = proved; N = numerical evidence.

Paper	Central output	Status
Q1	Phase coherence; $E = -\hat{a}\cdot\hat{b}$; Tsirelson; Born rule ($j = \frac{1}{2}$)	P
Q1 numerical	Rank $W_n^{(c)} = 0$ for $q = 29$, 4 conjugate pairs	N
Q2	Co-admissibility; SU(2) stable fixed point (Thm 8.6)	P
Q3	Proto-state = singlet (all j); universal correlator; Born rule (all j)	P
Bell	Bell factorizability inapplicable in non-injective frameworks	P

5 Inputs from Upstream Results

1. **$F_n \simeq V_\rho$ and Heisenberg structure** (Note 5): the identification of the admissible fibre as the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and the non-trivial commutator $[X,Y] = Z$ are the algebraic prerequisites for the phase coherence argument.

2. **Admissibility thread and SU(2) sector** (Note 1): the thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$ and the spin-$\frac{1}{2}$ identification (O29) specify the arena for the quantum structure derivation.

3. $\mathrm{Im}\,\mathbb{H} \cong \mathfrak{su}(2)$**; isotropic saturation** (O23 [11]): the isotropy condition $M \propto I$ and the identification of the three internal directions as $\mathrm{Im}\,\mathbb{H}$ are essential inputs to the correlator derivation.

4. **Parity involution and unit-norm amplitudes** (O18 [5]; O22 [9]): the parity involution $c \leftrightarrow q - c$ provides the normalisation of the correlator; O22 ensures unit-norm amplitudes at saturation.

5. **BI indiscernibility** (BornInfeld [3]): the BI evenness $\mathcal{S}_{\mathrm{BI}}[-\chi] = \mathcal{S}_{\mathrm{BI}}[\chi]$ forces the minimal fibre condition and underpins the indiscernibility of conjugate Weil blocks used in Q1.

6. **Non-injectivity as structural necessity** (ENI [7]): the Bell paper's non-applicability argument rests directly on the non-injectivity theorem of ENI: any genuinely emergent description has $S_\Pi > 0$, which is what prevents Bell factorization.

6 Outputs to Downstream Branches

7 Status of Results

7.1 Proved results

1. *Phase coherence* (Q1 Theorem 2.7): admissibility preserves BI indiscernibility; metaplectic phase coherence is a structural consequence.

Table 2: Principal outputs and downstream consumers.

Output	Consumer	Status
Phase coherence; coherent amplitude structure	Note 6 (fermions), Q-series	P
$E(\hat{a},\hat{b}) = -\hat{a}\cdot\hat{b}$ (spin-$\frac{1}{2}$)	Quantum phenomenology	P
Tsirelson bound $2\sqrt{2}$	Foundations applications	P
Born rule in SU(2) sector	Quantum measurements	P
SU(2) as stable fixed point	Note 1 (admissibility thread)	P
Universal correlator (all admissible j)	Quantum applications	P
Bell factorizability non-applicability	Quantum foundations	P
Observable signature: rank $W_n^{(c)} = 0$	O-series validation	N

2. *Observable rank signature* (Q1 Corollary 2.9): rank inflation detects any coherence violation; falsifiable.

3. *Singlet correlator at spin-$\frac{1}{2}$* (Q1 Theorem 2.14): derived from O18, O23, Q1 coherence; no quantum postulate.

4. *Tsirelson bound* (Q1 Corollary 2.16): $|S_{\mathrm{CHSH}}| \leq 2\sqrt{2}$; unconditional.

5. *Born rule at spin-$\frac{1}{2}$* (Q1 Theorem 2.18): unique assignment compatible with structural constraints; no Gleason theorem needed.

6. *Co-admissibility* $\lambda_{1/2} = \lambda_{3/2} = 18$ (Q2): proved from the spectral data of $2I$.

7. SU(2) *as unique stable fixed point* (Q2 Theorem 8.6): the admissibility flow selects $j = \frac{1}{2}$ in the LPS limit $p \to \infty$.

8. *Proto-state is the singlet for all j* (Q3 Theorem 4.1): Schur's lemma on the Clebsch–Gordan decomposition; BI indiscernibility.

9. *Universal spin-j correlator* (Q3 Theorem 5.2): $E(\hat{a},\hat{b}) = -\frac{j(j+1)}{3}(\hat{a}\cdot\hat{b})$ for all five admissible sectors.

10. *Born rule for all admissible j* (Q3 Corollary 6.1): follows from the universal correlator and the uniqueness argument of Q1 Theorem 2.18.

11. *Bell factorizability non-applicable* (Bell paper [2]): proved from non-injectivity and informational completeness of Π; published in *Quantum Reports*.

7.2 Numerical evidence

1. *Rank* $W_n^{(c)} = 0$ (Q1): structural indistinguishability confirmed for $q = 29$ across all four conjugate pairs throughout the admissible regime. Extension to $q \in \{61, 101, 151, 211, 307, 401\}$ is identified as a validation target.

7.3 Open hypotheses

1. *Born rule for general observables.* Theorem 2.18 of Q1 and Corollary 6.1 of Q3 derive the Born rule for the SU(2) parity sector. Extension to arbitrary observables (beyond spin-$\frac{1}{2}$ and the five admissible sectors of $2I$), multipartite systems, and continuous spectra requires extending the fibre structure beyond the two-element parity case.

2. *Admissibility flow beyond* SU(2). Q2 Theorem 8.6 establishes SU(2) as the stable fixed point within the co-admissible family on $2I$. Whether the flow has additional fixed points corresponding to SU(3) requires identifying the analogue of the $\lambda_{1/2} = \lambda_{3/2}$ co-admissibility condition for a group with SU(3) structure, connecting to the O-series.

3. *Multipartite entanglement.* The present derivations are for bipartite systems in the SU(2) sector. The co-admissibility of spin-$\frac{1}{2}$ and spin-$\frac{3}{2}$ on $2I$ raises the question of entangled states mixing both sectors. A multipartite extension beyond the single-pair setting is open.

8 Open Deliverables

Two operational deliverables define the boundary of the sub-programme. They subsume the three open hypotheses listed in Section 7.3.

Open deliverable 1: Born rule for general observables. This is the primary remaining open direction of the quantum bridge programme. The SU(2) case is complete. The general case requires either an extension of the parity-sector argument to arbitrary fibre structures, or a direct Gleason-type theorem for the admissible measure on V_ρ.

Open deliverable 2: Numerical validation across the full prime range. The rank-$W_n^{(c)} = 0$ signature of phase coherence has been confirmed for $q = 29$. Systematic verification across $q \in \{29, 61, 101, 151, 211, 307, 401\}$ would provide a comprehensive empirical confirmation and allow tracking of the boundary transition (rank-1 residual at saturation) as a function of q.

9 Conclusion

The quantum structure sub-programme derives the structures of quantum mechanics from the admissibility constraints of A1–A4, without importing quantum postulates.

Phase coherence is forced by BI indiscernibility of conjugate Weil blocks (Q1). The singlet correlator $E = -\hat{a} \cdot \hat{b}$ and the Tsirelson bound $2\sqrt{2}$ follow from four independently established inputs (Q1). The Born rule is the unique probability assignment compatible with the structural constraints (Q1). SU(2) is the unique stable fixed point of the admissibility flow (Q2). All results extend to the universal spin-j case for all five admissible sectors of $2I$ (Q3). Bell factorizability does not apply in non-injective effective descriptions (Bell paper, published in *Quantum Reports*).

The sub-programme is complete within the SU(2) sector. The primary open problem — the Born rule for general observables beyond SU(2) — requires an extension of the fibre structure that the current programme does not yet have.

References

[1] J. Beau. Admissible Non-Injective Transitions as the Primitive of Physical Description, 2026. Cosmochrony Foundation Paper "M", doi:10.5281/zenodo.19616329.

[2] J. Beau. Bell-Inequality Violations from Non-Injective Projection, 2026. Cosmochrony companion paper B, doi:10.5281/zenodo.18371173.

[3] J. Beau. Bounded Relaxation and the Dynamical Selection of Spacetime Geometry, 2026. Cosmochrony companion paper C about Born-Infeld, doi:10.5281/zenodo.18407505.

[4] J. Beau. From Admissibility to Quantum Structure: Phase Coherence and Correlations from Non-Injective Projection, 2026. spectral admissibility subprogram "Q1" paper, doi:10.5281/zenodo.19561060.

[5] J. Beau. Minimal Fibre Structure of the Non-Injective Projection from Born–Infeld Indiscernability: Derivation of the Parity Involution, 2026. Spectral admissibility sub-program "O18" paper, doi:10.5281/zenodo.19321413.

[6] J. Beau. Non-Factorisability and the Emergence of Heisenberg Structure from Admissibility Constraints, 2026. spectral admissibility subprogram paper, doi:10.5281/zenodo.19635395.

[7] J. Beau. Non-Injectivity as a Structural Necessity of Genuine Emergence: A No-Go Theorem for Faithful Emergent Descriptions, 2026. Cosmochrony companion paper "ENI" or L, doi:10.5281/zenodo.19391746.

[8] J. Beau. Quantum Structure beyond Spin-1/2: Co-Admissible Sectors and the Emergence of SU(2) as a Fixed Point Authors/Creators, 2026. spectral admissibility subprogram "Q2" paper, doi:10.5281/zenodo.19616444.

[9] J. Beau. Shell-Alignment from Projection Locking: A Discrete Admissibility Theorem under Born-Infeld Saturation, 2026. Spectral admissibility sub-program "O22" paper, doi:10.5281/zenodo.19367375.

[10] J. Beau. Spectral Admissibility under Bounded Flux: Non-Abelian Mode Selection on Cayley Graphs of SU(2) Subgroups, 2026. Step A of Cosmochrony Spectral Admissibility sub-Program, doi:10.5281/zenodo.18944985.

[11] J. Beau. Three Stable Directions from Quaternionic Minimality: Derivation of $\Sigma_c(n_3) = 3$ from Born–Infeld Fibre Admissibility, 2026. Spectral admissibility sub-program "O23" paper, doi:10.5281/zenodo.19375136.

[12] Jérôme Beau. Universal Spin-j Singlet from Admissibility: Completion of the SU(2) Quantum Sector, 2026. "Q3" paper, doi:10.5281/zenodo.19979535.